

Solving Basic Equations & Integer Operations

Pre-Algebra Worksheet · Grade 6–8

Name: _____

Date: _____

Learning Objectives

- Add and subtract integers using number line reasoning
- Identify and apply additive and multiplicative inverses
- Solve one-step and two-step algebraic equations

Problems

1. Solve the subtraction problem using the number line. Start at 7 and move 2 units in the negative direction.

$$7 - 2 = ?$$

2. Solve the subtraction problem. Start at 3 and move 8 units in the negative direction. Be careful to include the correct sign in your answer.

$$3 - 8 = ?$$

3. Use the number line concept to solve the following subtraction. Start at 9 and move 11 units in the negative direction.

$$9 - 11 = ?$$

4. Solve the following problem involving negative integers. Start at negative 2 and move 5 units further in the negative direction.

$$-2 - 5 = ?$$

5. Find the additive inverse of 7. The additive inverse is the number you must add to 7 so that the result equals 0.

$$7 + ? = 0$$



6. Find the multiplicative inverse of 4. The multiplicative inverse is the number you must multiply by 4 so that the result equals 1.

$$4 \times ? = 1$$

7. Solve the one-step equation for x by using the concept of additive inverse.

$$x + 5 = 12$$

8. Solve the one-step equation for x by using the concept of multiplicative inverse.

$$3x = 21$$

9. Solve the two-step equation for x. First apply the additive inverse, then the multiplicative inverse.

$$2x - 6 = 10$$

10. Solve the two-step equation for x involving negative integers. Apply both additive and multiplicative inverse properties.

$$-4x + 3 = -13$$



Solving Basic Equations & Integer Operations — Answer Key

Pre-Algebra Worksheet · Grade 6–8

Answer Key

1. Answer: 5

- Start at 7 on the number line.
 - Move 2 units to the left (negative direction): $7 \rightarrow 6 \rightarrow 5$.
 - Answer: 5
-

2. Answer: -5

- Start at 3 on the number line.
 - Move 8 units to the left: $3 \rightarrow 2 \rightarrow 1 \rightarrow 0 \rightarrow -1 \rightarrow -2 \rightarrow -3 \rightarrow -4 \rightarrow -5$.
 - Answer: -5
-

3. Answer: -2

- Start at 9 on the number line.
 - Move 11 units to the left: 9 crosses 0 and reaches -2.
 - Answer: -2
-

4. Answer: -7

- Start at -2 on the number line.
 - Move 5 more units to the left: $-2 \rightarrow -3 \rightarrow -4 \rightarrow -5 \rightarrow -6 \rightarrow -7$.
 - Answer: -7
-

5. Answer: -7

- The additive inverse of a number n is $-n$.
 - So the additive inverse of 7 is -7.
 - Check: $7 + (-7) = 0$ ✓
-

6. Answer: $1/4$

- The multiplicative inverse (reciprocal) of a number n is $1/n$.
 - So the multiplicative inverse of 4 is $1/4$.
 - Check: $4 \times (1/4) = 1$ ✓
-

7. Answer: $x = 7$

- To isolate x , add the additive inverse of 5 (which is -5) to both sides.
 - $x + 5 + (-5) = 12 + (-5)$
 - $x = 7$
-

8. Answer: $x = 7$

- To isolate x , multiply both sides by the multiplicative inverse of 3, which is $1/3$.
-





- $(1/3) \times 3x = 21 \times (1/3)$
 - $x = 7$
-

9. Answer: $x = 8$

- Step 1: Add the additive inverse of -6 (which is +6) to both sides: $2x - 6 + 6 = 10 + 6 \rightarrow 2x = 16$.
 - Step 2: Multiply both sides by the multiplicative inverse of 2 (which is $1/2$): $x = 16 \times (1/2)$.
 - $x = 8$
-

10. Answer: $x = 4$

- Step 1: Add the additive inverse of 3 (which is -3) to both sides: $-4x + 3 - 3 = -13 - 3 \rightarrow -4x = -16$.
 - Step 2: Multiply both sides by the multiplicative inverse of -4 (which is $-1/4$): $x = -16 \times (-1/4)$.
 - $x = 4$
-

